

Mathematics A

Unit 2 • Chapter OL-T40 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

1 classified items • 6 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T40 • Expertise Q16+ • 1 items • 6 marks

Chapter	Items	Marks	Starts
01. OL-T40 Angles in Polygons & Parallel Lines	1	6	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Angles in Polygons & Parallel Lines

Unit 2 • Expertise • 6 marks • 1 item(s)

June 2022 | 4MA1-1H Q23 | 6 marks

Solve an irregular polygon angle or side-count problem

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q23 Q23

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q23 Q23

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

Chapter: Angles in Polygons & Parallel Lines

6 marks

Q23 – Chapter: Angles in Polygons & Parallel Lines

Family: Use regular and general polygon angles • Variant: Solve an irregular polygon angle or side-count problem

Method

The interior-angle sequence has first term 177 and common difference $d = -2$.

Why: Successive listed angles decrease by 2 degrees.

Method

Its sum is $S_n = n/2[2(177) + (n - 1)(-2)]$.

Why: Apply the arithmetic-series sum formula.

Method

The interior-angle sum of an n -sided polygon is $180(n - 2)$, so $n/2[354 - 2(n - 1)] = 180(n - 2)$.

Why: Equate the sequence sum to the polygon angle sum.

Method

Simplifying gives $n^2 + 2n - 360 = 0$.

Why: Expand and collect all terms on one side.

Method

$n^2 + 2n - 360 = (n - 18)(n + 20)$, so $n = 18$ or $n = -20$.

Why: Factorise the quadratic and use the zero-product rule.

Method

The number of sides is positive and $n > 5$, hence $n = 18$.

Why: Apply the domain condition for a polygon.

Final answer

number of sides: $n = 18$